

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and implemented **multi-agent orchestrator/specialist workflow patterns** using **Python** and **Spring Boot REST APIs** to decompose complex payment-processing logic into reusable LLM-driven agents, reducing manual intervention by **40%** and enabling real-time SLA compliance across banking transaction flows.
- Applied **context engineering techniques** including **system prompts, agent customization, and agent skills configuration** to fine-tune LLM reasoning for financial-domain workflows; evaluated LLM context window management across **ISO 20022 message parsing** and **SWIFT MT/MX transaction validation**, achieving **95% accuracy** in production.
- Built and evaluated **agentic framework prototypes** using **LangGraph** and **CrewAI** to orchestrate multi-step workflows for cross-border payments, reconciliation, and regulatory compliance; conducted **prompt engineering evaluation and red-teaming** to stress-test agent robustness across ambiguous financial-services edge cases.
- Implemented **TDD, CI/CD pipelines (Jenkins), and code-review practices** across **Python, Java, and Node.js** agent implementations; integrated agents with **AWS Lambda, CloudWatch, and Docker** containerization to deploy experimental AI-native microservices with automated testing and fast iteration cycles.
- Developed **data pipelines and ETL workflows** using **Python (pandas, NumPy)** and **SQL (PostgreSQL, Oracle)** to ingest, transform, and contextualize financial transaction data for agent knowledge bases; built **Power BI dashboards** to monitor agent performance metrics, output quality, and cost efficiency across **150+ daily agentic workflow executions**.
- Collaborated with architecture and engineering teams to design **next-generation AI-native platforms** using **leading-edge LLM tools**; demonstrated bias for experimentation and rapid prototyping by evaluating AI outputs critically, documenting lessons learned, and iterating on agent designs to optimize for business outcomes in fast-moving, ambiguous AI tooling landscape.

SKILLS

Programming Languages & Software Engineering – Python, Java, Node.js, REST APIs, Spring Boot, Flask, FastAPI, TDD, CI/CD (Jenkins, Git).

Agentic AI & LLM Engineering – Multi-agent orchestrator/specialist patterns, LangGraph, CrewAI, AutoGen, context engineering, system prompts, agent customization, agent skills, context window management, prompt engineering evaluation, red-teaming, LLM reasoning.

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, containerization, Linux, shell scripting, change-control management.

Data Engineering & Analytics – SQL (PostgreSQL, MySQL, Oracle), ETL, data pipelines, pandas, NumPy, scikit-learn, Power BI (dashboards, DAX, data modelling), Tableau, data warehousing.

Financial Services Domain – SWIFT messaging (MT/MX formats), ISO 20022 message flows, SEPA, instant payments, cross-border payments, payments operations, PSD2, Open Banking, reconciliation workflows, regulatory compliance (GDPR, data protection).

Enterprise Systems & Monitoring – ERP integration support, Splunk, Dynatrace, operational alerting, production support, incident response, SLA management.

Collaboration & Documentation – Jira, Confluence, Agile/Scrum, cross-functional stakeholder engagement, UAT support, requirements gathering.

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin	02/2026
BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India	01/2021

PROJECTS

Agentic Payments Orchestration Platform (HCL Technologies)

- Architected a **multi-agent agentic workflow system** using **LangGraph and Python** to orchestrate specialist agents for payment validation, regulatory compliance checking, and cross-border settlement routing; agents leveraged **system prompts and ISO 20022 context engineering** to parse and reason over complex payment message structures, reducing human review time by **60%**.
- Implemented **context window management and prompt engineering evaluation** strategies to optimize LLM token usage and accuracy across **SWIFT MT/MX and ISO 20022 message transformation**; conducted **red-team testing** against edge cases (malformed messages, regulatory boundary conditions) achieving **98% agent output correctness** in production.
- Deployed agents as **containerized microservices (Docker)** on **AWS Lambda** with **CloudWatch monitoring, CI/CD pipelines (Jenkins)**, and **automated unit + integration tests**; designed fallback patterns to gracefully degrade to human operators during agent uncertainty, maintaining **99.8% SLA uptime** for payment-operations dashboards.
- Built **Power BI dashboards** integrating **SQL queries (PostgreSQL)** to monitor agent performance KPIs: throughput (**2,400+ payments/day**), latency (**avg 1.2s end-to-end**), cost per transaction, and LLM API consumption; enabled real-time visibility for banking domain business analysts and operations teams.
- Documented agent design patterns, prompt templates, and context engineering best practices in **Confluence wikis** for cross-functional teams; facilitated knowledge-sharing sessions with architecture engineers on lessons learned from **ambiguous AI tooling landscape**, including evaluation of **CrewAI and AutoGen alternatives** for future platform iterations.

SWIFT Payments Operations Dashboard & Compliance Monitoring (HCL Technologies)

- Engineered production-support and real-time monitoring systems for **SWIFT MT/MX messaging platform** using **Splunk and Dynatrace**, delivering operational alerting and transaction-flow visibility across **cross-border and SEPA instant payments** workflows; reduced P1 incident detection time by **45%** and enabled SLA compliance reporting for banking client regulators.
- Developed **Python ETL pipelines and SQL analytics queries (PostgreSQL, Oracle)** to validate payment message structure, reconcile transaction ledgers, and audit GDPR-compliant data flows; integrated **REST API calls** to interconnected ERP and core banking systems, automating **daily reconciliation cycles** that historically required **8 hours of manual effort**.
- Collaborated with banking-domain BAs and compliance teams to encode **ISO 20022 message validation rules, PSD2 regulatory requirements, and data-protection mandates** into automated monitoring logic; supported UAT cycles and post-go-live triage, ensuring **100% audit-trail completeness** for regulatory reporting.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023

- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)